

Majorization-based Evolutionary Algorithm for Balanced System Optimization [★]

Zhaobo Liu^{*}, Tao Zeng^{**}, Yanfang Mo^{***},
Miaomiao Wang^{****}, Wenjing Hong^{**}, Zexuan Zhu^{**}

^{*} *Institute for Advanced Study, Shenzhen University, Shenzhen, China
(e-mail: liuzhaobo@szu.edu.cn).*

^{**} *School of Artificial Intelligence, Shenzhen University, Shenzhen,
China (e-mail: szuztao@163.com; {hongwj, zhuzx}@szu.edu.cn).*

^{***} *Division of Industrial Data Science, School of Data Science,
Lingnan University, Hong Kong SAR, China (e-mail:
yanfangmo@LN.edu.hk).*

^{****} *Department of Electrical Engineering, City University of Hong
Kong, Hong Kong SAR, China (e-mail:
miaomiao.wang@cityu.edu.hk).*

Abstract:

Complex system optimization problems often require solutions that are both efficient and balanced, a challenge naturally arising in homogeneous vector-based performance evaluation. Traditional multi-objective evolutionary algorithms typically address this issue by introducing problem-specific balance metrics, which limits their general applicability. This paper proposes Majorization-NSGA-II (M-NSGA-II), an algorithm that redefines solution ranking by replacing Pareto dominance with the weak majorization preorder. This selection mechanism provides a principled preference for balanced solutions and is theoretically justified for aggregate performance criteria that are Schur-convex and monotonically increasing. Through a multi-UAV path-planning case study, we show that M-NSGA-II discovers solutions that are simultaneously more efficient, more balanced, and more robust in terms of worst-agent cost. These results indicate that majorization-based ranking is an effective framework for balanced optimization in homogeneous multi-component systems.

Keywords: Multi-objective optimization, Evolutionary algorithms, Majorization, System robustness, Fairness, Multi-agent systems.

1. INTRODUCTION

Many complex multi-component systems require solutions that are both efficient and balanced across identical entities, including cloud platforms (Mishra et al., 2020; Kumar and Kumar, 2019; Devi et al., 2024), finance (Bakry et al., 2021; Chiam et al., 2008; Cesarone et al., 2020), and robotic teams (Amir et al., 2025; Bettini et al., 2025; Harmati and Skrzypczyk, 2009). In such problems, a solution is naturally evaluated by a homogeneous vector whose entries describe the costs, loads, or risks assigned to individual components. Standard multi-objective evolutionary algorithms (MOEAs) such as NSGA-II (Deb et al., 2002) can optimize this vector, but Pareto dominance is often

indifferent to balance; for example, (10, 10) and (1, 19) are mutually non-dominating.

This setting is different from general MOO with heterogeneous objectives such as cost, safety, and energy. Here, all objectives have the same physical meaning but are associated with different system components. The main conflict is therefore not a trade-off among different criteria, but an allocation trade-off among identical agents or resources. This distinction is important because it permits the use of order relations that explicitly compare the spread of objective vectors.

Existing approaches typically introduce a specific balance metric, such as variance, either for post-processing or as an additional objective (Wang et al., 2025). This is useful but metric-dependent. Inspired by majorization order in load balancing and security (Mo et al., 2017, 2024; Mo and Qiu, 2024), this paper proposes Majorization-NSGA-II (M-NSGA-II), which replaces Pareto dominance in the ranking step with weak majorization. The contributions are: (i) a simple NSGA-II variant that embeds a balance preference for homogeneous cost vectors; (ii) a clarified theoretical scope based on monotone Schur-convex preferences; and (iii) a multi-UAV path-planning study showing

[★] This work was supported in part by the National Natural Science Foundation of China under Grant 12401585; in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2024A1515011542; in part by the General Program of Shenzhen Natural Science Foundation under Grant JCYJ20250604181037012; in part by Lingnan University under Grant DR25E7 and Grant DR26F4; and in part by the Hong Kong RGC under Projects CityU 11207823 and CityU 11206024.

Corresponding author: yanfangmo@LN.edu.hk

statistically significant gains in efficiency, balance, and worst-agent cost.

2. THEORETICAL FOUNDATION

2.1 Problem Formulation

Let x be a candidate solution from the feasible solution space Ω . The performance of this solution is described by a real-valued vector of n components:

$$F(x) = (f_1(x), f_2(x), \dots, f_n(x)), \quad (1)$$

where $f_i(x)$ is the cost, load, or risk of component i . The objectives are homogeneous and minimized simultaneously; the goal is not only Pareto efficiency but also balance among components.

2.2 Majorization and Schur-Convexity

Majorization is a preorder for comparing vector spread.

Definition 2.1. (Majorization). Let $v = (v_1, \dots, v_n)$ and $w = (w_1, \dots, w_n)$ be two vectors. Let $v_{[k]}$ and $w_{[k]}$ denote the k -th largest element of each vector. The vector v is said to be majorized by w , denoted as $v \prec w$, if the following conditions hold:

$$\sum_{k=1}^j v_{[k]} \leq \sum_{k=1}^j w_{[k]}, \quad \forall j = 1, \dots, n-1 \quad (2)$$

$$\sum_{k=1}^n v_{[k]} = \sum_{k=1}^n w_{[k]}. \quad (3)$$

If $v \prec w$, then v is less spread out than w , e.g., $(4, 4, 4) \prec (3, 5, 4) \prec (2, 6, 4)$. Since the equality constraint on the total sum is restrictive when costs are not conserved, we use weak majorization.

Definition 2.2. (Weak Majorization). The vector v is said to be weakly majorized by w , denoted $v \prec_w w$, if only the partial sum condition holds:

$$\sum_{k=1}^j v_{[k]} \leq \sum_{k=1}^j w_{[k]}, \quad \forall j = 1, \dots, n. \quad (4)$$

Weak majorization captures both efficiency and balance in a single preorder. For example, after sorting components in descending order, $(4, 4, 4) \prec_w (6, 3, 3)$ because the largest partial sum and all subsequent partial sums are no larger. The first vector is therefore preferable not only because it is more even, but also because no prefix of the largest burdens is worse. In contrast, vectors such as $(5, 5, 1)$ and $(7, 2, 2)$ may be incomparable: one has a smaller largest component, while the other can have a smaller cumulative burden after several components. This incomparability is important in evolutionary search because it preserves multiple plausible trade-offs instead of forcing all solutions into a single scalar ranking.

Majorization is closely connected to Schur-convex functions.

Definition 2.3. (Schur-Convexity). A function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be Schur-convex if for any two vectors $v, w \in \mathbb{R}^n$, $v \prec w$ implies $g(v) \leq g(w)$. Conversely, a function is Schur-concave if $v \prec w$ implies $g(v) \geq g(w)$.

Schur-convex functions penalize imbalance; examples include variance and the sum of squares. A classical result states that v is majorized by w if and only if $g(v) \leq g(w)$ for all continuous Schur-convex functions g (Marshall et al., 2011).

While Schur-convexity alone does not guarantee order preservation under weak majorization, the combination with monotonicity does. The following key proposition (Marshall et al., 2011, Theorem A.8), connects these concepts.

Definition 2.4. Let $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function. The function g is monotonically increasing if $v_i \leq w_i$ for all $i = 1, \dots, n$ implies $g(v) \leq g(w)$.

Proposition 2.5. A real-valued continuous function g preserves weak majorization, i.e., $v \prec_w w$ implies $g(v) \leq g(w)$, **if and only if** g is both Schur-convex and monotonically increasing.

2.3 The Target Problem Class for Balanced Optimization

Proposition 2.5 justifies majorization-based selection for problems whose aggregate preference $g(F)$ is both Schur-convex and monotonically increasing; denote this class by $\mathcal{P}_{\text{SCMI}}$. It includes common robustness and fairness objectives, such as minimax cost $g(F) = \max_i f_i$ and convex load penalties $g(F) = \sum_i f_i^2$ (Jain et al., 1984; Ebeling, 2019). These assumptions are also practical applicability conditions. A problem is a suitable candidate when:

- the objective components are homogeneous and comparable, such as path lengths of identical UAVs, loads of parallel servers, or costs assigned to identical agents;
- smaller values are preferred componentwise, so that increasing any component cannot improve the aggregate system performance;
- the desired aggregate preference penalizes uneven distributions, for example through maximum cost, sum of convex component penalties, or related load-balancing criteria.

The method is less appropriate for heterogeneous objectives with different physical meanings, such as cost, safety, and energy, unless they are transformed into comparable componentwise costs. Thus weak majorization is a justified ranking tool within $\mathcal{P}_{\text{SCMI}}$, while problems outside this class require additional modeling choices or alternative dominance relations.

The class $\mathcal{P}_{\text{SCMI}}$ covers several engineering cases. In “weakest-link” systems, reliability or mission completion is constrained by the most burdened component, so minimizing $\max_i f_i$ is natural. In load balancing, convex penalties such as $\sum_i f_i^2$ discourage concentrating work on a few components. In multi-agent systems, the same structure appears when individual agents are interchangeable but unequal workload creates robustness or fairness concerns. These examples show that the assumptions are restrictive in a useful way: they identify when balance is part of the system preference rather than an arbitrary post-processing score.

For practical use, the following verification procedure can be applied before using M-NSGA-II. First, normalize or

Algorithm 1. Majorization-based Sorting

1: **Input:** Population R_t ; **Output:** fronts $F = \{F_1, F_2, \dots\}$.
2: $F \leftarrow \emptyset$, $k \leftarrow 1$.
3: **while** R_t is not empty **do**
4: $F_k \leftarrow \emptyset$.
5: **for** each individual $p \in R_t$ **do**
6: **if** no $q \in R_t$ satisfies $q \succ_M p$ **then**
7: $F_k \leftarrow F_k \cup \{p\}$.
8: **end if**
9: **end for**
10: $F \leftarrow F \cup \{F_k\}$; $R_t \leftarrow R_t \setminus F_k$; $k \leftarrow k + 1$.
11: **end while**

model the objectives so that all components represent the same type of cost. Second, confirm that increasing any component cannot improve the decision maker's aggregate preference. Third, check whether the intended preference penalizes dispersion; this is usually true for worst-component objectives, sums of convex component penalties, and many load-balancing criteria. If these checks fail, weak majorization may still be useful as a heuristic, but the order-preservation guarantee in Proposition 2.5 no longer applies directly.

3. THE MAJORIZATION-NSGA-II ALGORITHM

M-NSGA-II follows the elitist loop of NSGA-II (Deb et al., 2002). In each generation, a child population Q_t is generated from the parent population P_t using the same variation operators, and $R_t = P_t \cup Q_t$ is ranked to select P_{t+1} . The only structural change is that Pareto sorting is replaced by majorization-based sorting.

This design keeps the algorithmic novelty deliberately focused. Variation, elitism, and diversity preservation remain those of NSGA-II, so differences in performance can be attributed mainly to the modified ranking relation. This also makes the method easy to incorporate into existing MOEA implementations.

3.1 Majorization-based Sorting

The sorting procedure partitions R_t into fronts $F_1, F_2, \dots$ using weak majorization. For minimization, a smaller and more balanced cost vector is preferred:

Definition 3.1. (Dominance via Weak Majorization). For a minimization problem, a solution x_i is said to majorization-dominate a solution x_j , denoted $x_i \succ_M x_j$, if and only if its objective vector $F(x_i)$ is weakly majorized by the objective vector $F(x_j)$:

$$x_i \succ_M x_j \iff F(x_i) \prec_w F(x_j) \quad (5)$$

where $\prec_w$ is the weak majorization relation from Definition 2.2.

By Proposition 2.5, this ranking is order-preserving for $\mathcal{P}_{\text{SCMI}}$. Algorithm 1 repeatedly extracts the current majorization non-dominated set, removes it, and continues until all solutions are ranked.

3.2 Complexity and Search Dynamics

Let $P = |R_t|$ and let m be the number of objectives. A direct majorization comparison sorts objective

vectors and checks cumulative sums, giving $O(m \log m)$ time per comparison and $O(P^2 m \log m)$ per sorting step. Caching sorted vectors and cumulative sums reduces this to $O(Pm \log m + P^2 m)$, so the additional cost over Pareto sorting mainly comes from vector sorting. Weak majorization can also leave many solutions incomparable, especially for large m ; hence M-NSGA-II keeps crowding distance to preserve diversity. This provides selection pressure toward balanced low-cost regions but does not constitute a convergence proof.

Table 1. Sorting cost per generation for population size P and objective dimension m .

Ranking rule	Direct comparison	With cached vectors
Pareto sorting	$O(P^2 m)$	$O(P^2 m)$
Majorization sorting	$O(P^2 m \log m)$	$O(Pm \log m + P^2 m)$

Table 1 shows that the proposed ranking has the same quadratic dependence on population size as pairwise Pareto sorting, but introduces a sorting-related overhead in the objective dimension. In the UAV experiments, m is the number of UAVs and remains moderate. For very large m , both the computational overhead and the number of incomparable solutions may increase, which motivates future work on scalable approximations or partial-sum caching strategies.

To complement the complexity analysis, wall-clock runtime was measured over 30 runs with matched random seeds. In S1 and S2, M-NSGA-II required 19.489 ± 2.523 s and 34.468 ± 3.036 s, respectively, whereas NSGA-II required 19.238 ± 2.267 s and 33.270 ± 3.110 s. The ratios of M-NSGA-II runtime to NSGA-II runtime were 1.013 and 1.036, indicating modest overhead in the tested instances. For much larger m , the discriminative power of weak majorization may decline because fewer vectors dominate one another; additional preference information, adaptive aggregation, or hierarchical grouping may then be needed.

The stricter partial order removes solutions that are simultaneously large and unbalanced, while leaving many incomparable solutions. The crowding-distance step is therefore important for maintaining a spread of alternatives rather than collapsing to a single balanced point.

3.3 Population Update and Diversity Preservation

After sorting R_t , fronts are inserted into P_{t+1} in rank order. If the next front would exceed the population size, its members are selected by the standard NSGA-II crowding distance (Deb et al., 2002), which helps preserve diversity among mutually incomparable solutions.

4. EXPERIMENTAL VALIDATION

We evaluate M-NSGA-II on a multi-UAV path-planning problem, focusing on whether majorization-based ranking improves efficiency, balance, and worst-agent cost simultaneously.

4.1 Case Study: Multi-UAV Path Planning

Problem Formulation We consider a fleet of N UAVs starting from a depot S and visiting M targets, each of which must be served by exactly one UAV.

The performance of a path plan x is described by the vector of individual travel distances for each UAV: $F(x) = (f_1(x), \dots, f_N(x))$. The cost $f_i(x)$ for UAV U_i is its total Euclidean flight distance. Given an ordered path $P_i = (T_{i,1}, \dots, T_{i,k_i})$ assigned to U_i , its cost is given by:

$$f_i(x) = \begin{cases} 0, & P_i = \emptyset, \\ d(S, T_{i,1}) + \sum_{j=1}^{k_i-1} d(T_{i,j}, T_{i,j+1}) + d(T_{i,k_i}, S), & P_i \neq \emptyset. \end{cases}$$

where $d(A, B)$ is the Euclidean distance.

The benchmark uses an obstacle-free Euclidean setting to isolate workload allocation and route ordering. Obstacle-rich environments can be incorporated by replacing Euclidean distances with shortest feasible path lengths or a precomputed distance matrix; the objective vector remains homogeneous as long as each component represents the same type of travel cost.

Optimization Objectives We seek solutions with:

- **Efficiency:** collectively low path lengths, corresponding to low fuel consumption or operational time.
- **Balance:** similar path lengths across UAVs, avoiding premature battery depletion or unequal mission completion times.

Decision Encoding Each solution is encoded as $\mathbf{x} = [\boldsymbol{\pi}, \mathbf{s}]$, where $\boldsymbol{\pi}$ is a target permutation and $\mathbf{s} \in \{1, \dots, N\}^M$ assigns each target to a UAV. The route of UAV k is decoded by extracting targets with $s_j = k$ while preserving their order in $\boldsymbol{\pi}$.

We use synchronized order crossover (Kora and Yadlapalli, 2017), copying target subsequences and their UAV labels together. Mutation applies swap, insertion, and inversion to $\boldsymbol{\pi}$ (Deep and Mebrahtu, 2011), and reallocates either a single target or a contiguous block in $\mathbf{s}$.

Experimental Setup Two scenarios are used: S1 with $N = 5$ UAVs and $M = 30$ targets, and S2 with $N = 10$ UAVs and $M = 50$ targets. Targets are sampled uniformly from the unit square and fixed for all algorithms; the depot is at the origin. Each method is run for 30 independent trials, with population size 100 and 200,000 function evaluations.

Performance Metrics for Evaluation We assess the final fronts using:

- **TD:** $\text{TD}(x) = \sum_i f_i(x)$, measuring efficiency.
- **VAR:** $\text{VAR}(x) = \text{Var}(f_i(x))$, a Schur-convex balance diagnostic.
- **MAX:** $\text{MAX}(x) = \max_i f_i(x)$, a monotone Schur-convex worst-agent metric.

For each run, we analyze both specialist solutions minimizing each metric and the average metric values over the final front. Significance is verified by the Wilcoxon rank-sum test at the 5% level.

These metrics serve different roles: TD measures total mission cost, MAX matches the monotone Schur-convex robustness criterion, and VAR diagnoses workload balance. Reporting all three tests whether the method im-

proves worst-agent behavior, efficiency, and pure balance simultaneously.

Competing Algorithms M-NSGA-II is compared with MOEA/D (Zhang and Li, 2007), NSGA-II (Deb et al., 2002), and NSGA-III (Deb and Jain, 2014). All methods optimize $F(x) = (f_1(x), \dots, f_N(x))$ and use the same population size, evaluation budget, target instances, encoding, initialization, crossover, mutation, and statistical test. MOEA/D neighborhoods and NSGA-III reference directions follow standard recommended settings and are not tuned separately for any method, avoiding benchmark-specific tuning.

For fairness, the standard baselines and M-NSGA-II all use the same componentwise path-length vector; the comparison therefore focuses on how their selection mechanisms organize the search over that common objective representation.

To address balance-aware alternatives, we further compare against three baselines implemented with the same population model and variation operators: NSGA-II-TVM, which optimizes $(\text{TD}(x), \text{VAR}(x), \text{MAX}(x))$ as augmented objectives; NSGA-II-Minimax, which optimizes $\text{MAX}(x)$; and NSGA-II-Scalarization, which optimizes $G_{\text{sca}}(x) = \text{TD}(x)/N + \text{MAX}(x)$. These baselines explicitly encode particular balance preferences, while M-NSGA-II ranks the original homogeneous vector by weak majorization.

4.2 Results and Discussion

For each run, we evaluate both specialist solutions selected by minimum TD, VAR, or MAX and the average quality of the final front. Against the standard MOEA baselines, M-NSGA-II obtains the best mean values in all specialist-solution comparisons for both scenarios, with significance verified by the Wilcoxon rank-sum test at the 5% level. Table 2 reports the strongest standard baseline for each specialist criterion, Table 3 retains all standard baseline rows for the comprehensive front analysis, and Table 4 summarizes the additional balance-aware comparisons.

Table 2. Specialist solutions. M-NSGA denotes M-NSGA-II; baseline row is the strongest baseline for the selected criterion; bold values are significant at the 5% level.

Case	Alg.	TD	VAR	MAX
S1-TD	NSGA-III	15.06±1.14	8.12±4.05	7.43±1.66
S1-TD	M-NSGA	9.23±0.58	1.06±0.38	2.93±0.22
S1-VAR	MOEA/D	20.99±1.47	0.036±0.021	4.45±0.29
S1-VAR	M-NSGA	12.08±0.29	0.021±0.018	2.61±0.03
S1-MAX	MOEA/D	18.45±1.00	0.134±0.089	4.13±0.16
S1-MAX	M-NSGA	11.78±0.41	0.084±0.092	2.60±0.02
S2-TD	NSGA-III	20.67±1.34	2.82±0.90	5.27±1.01
S2-TD	M-NSGA	19.54±0.96	0.56±0.20	2.82±0.18
S2-VAR	MOEA/D	35.17±1.82	0.098±0.027	4.01±0.17
S2-VAR	M-NSGA	22.57±0.60	0.061±0.044	2.61±0.08
S2-MAX	MOEA/D	32.07±1.15	0.176±0.078	3.78±0.10
S2-MAX	M-NSGA	22.13±0.77	0.124±0.081	2.58±0.07

The specialist results in Table 2 show that M-NSGA-II does not improve one metric by sacrificing the others. For example, when selecting the VAR-optimal solution, it obtains not only lower variance but also lower TD and MAX than the best baseline specialist solution. A similar pattern appears for TD-optimal and MAX-optimal

Table 3. Comprehensive final-front analysis. M-NSGA denotes M-NSGA-II; bold values are significant at the 5% level.

Scenario	Algorithm	TD	VAR	MAX
S1	MOEA/D	20.983 $\pm$ 0.145	3.917 $\pm$ 0.397	5.999 $\pm$ 0.123
S1	NSGA-II	21.520 $\pm$ 0.453	16.528 $\pm$ 1.449	11.241 $\pm$ 0.401
S1	NSGA-III	18.969 $\pm$ 0.525	7.273 $\pm$ 1.244	7.877 $\pm$ 0.433
S1	M-NSGA	10.738 $\pm$ 0.294	0.353 $\pm$ 0.116	2.756 $\pm$ 0.099
S2	MOEA/D	34.981 $\pm$ 0.154	1.552 $\pm$ 0.372	4.940 $\pm$ 0.126
S2	NSGA-II	37.291 $\pm$ 0.307	13.589 $\pm$ 0.856	12.965 $\pm$ 0.353
S2	NSGA-III	25.085 $\pm$ 1.548	4.053 $\pm$ 0.915	6.503 $\pm$ 0.650
S2	M-NSGA	21.039 $\pm$ 0.590	0.251 $\pm$ 0.073	2.696 $\pm$ 0.107

Table 4. Balance-aware baselines. M-NSGA denotes M-NSGA-II; TVM, Minimax, and Scalarization are selected by minimum VAR, MAX, and G_{sca} , respectively; bold values are significant at the 5% level.

Case	Alg.	TD	VAR	MAX
S1-TVM	M-NSGA	12.29 $\pm$ 0.25	0.0126 $\pm$ 0.0093	2.62 $\pm$ 0.04
S1-TVM	NSGA-TVM	14.09 $\pm$ 0.83	0.00016 $\pm$ 0.00015	2.84 $\pm$ 0.17
S2-TVM	M-NSGA	22.62 $\pm$ 0.85	0.0635 $\pm$ 0.0460	2.60 $\pm$ 0.12
S2-TVM	NSGA-TVM	27.09 $\pm$ 1.58	0.00267 $\pm$ 0.00150	2.79 $\pm$ 0.16
S1-Minimax	M-NSGA	12.01 $\pm$ 0.50	0.0646 $\pm$ 0.0773	2.61 $\pm$ 0.02
S1-Minimax	Minimax	13.04 $\pm$ 0.51	0.0054 $\pm$ 0.0088	2.67 $\pm$ 0.09
S2-Minimax	M-NSGA	22.22 $\pm$ 0.90	0.124 $\pm$ 0.074	2.56 $\pm$ 0.06
S2-Minimax	Minimax	25.96 $\pm$ 0.81	0.0215 $\pm$ 0.0234	2.72 $\pm$ 0.07
S1-Scalar	M-NSGA	9.32 $\pm$ 0.60	0.919 $\pm$ 0.226	2.77 $\pm$ 0.11
S1-Scalar	Scalar	9.93 $\pm$ 0.64	0.821 $\pm$ 0.339	2.73 $\pm$ 0.11
S2-Scalar	M-NSGA	20.13 $\pm$ 1.46	0.470 $\pm$ 0.293	2.62 $\pm$ 0.08
S2-Scalar	Scalar	20.03 $\pm$ 1.15	0.563 $\pm$ 0.266	2.67 $\pm$ 0.09

selections. This supports the central design goal: replacing Pareto ranking by weak majorization encourages balanced solutions without degrading overall route quality.

The front-level results in Table 3 further show that this effect is not limited to a few selected individuals. The fronts generated by M-NSGA-II have lower total cost, smaller dispersion, and lower worst-agent burden. The strong MAX performance aligns with the monotone Schur-convex preference class; the VAR improvement shows better pure balance, and the TD improvement suggests that pruning highly unbalanced solutions also helps concentrate search on efficient route structures.

The balance-aware comparisons in Table 4 show why directly optimizing a selected balance metric is not equivalent to majorization-based ranking. NSGA-II-TVM and the minimax baseline can obtain extremely small VAR values, but this often comes with higher TD and MAX, indicating an over-balancing tendency: path lengths become nearly equal while the total and worst-agent costs increase. The scalarized baseline is competitive on its own scalar criterion, yet its outcome depends on the particular scalar form and weighting. M-NSGA-II remains competitive across these evaluation rules and often achieves lower MAX, supporting the claim that weak majorization provides a robust efficiency-balance trade-off without pre-selecting one balance metric.

The comparison with NSGA-II is particularly informative because both algorithms share the same framework, encoding, and variation operators; the main difference is environmental selection. Thus, the performance gap is closely related to the majorization-based preference rather than

simply to the path encoding or mutation operators. This also calibrates the algorithmic contribution: M-NSGA-II is not a new evolutionary framework, but a targeted replacement of the ranking relation that changes the selection pressure in homogeneous objective spaces.

Visualization of Optimal Balanced Solutions Figure 1 visualizes the VAR-optimal solutions found by each algorithm for Scenario S1.

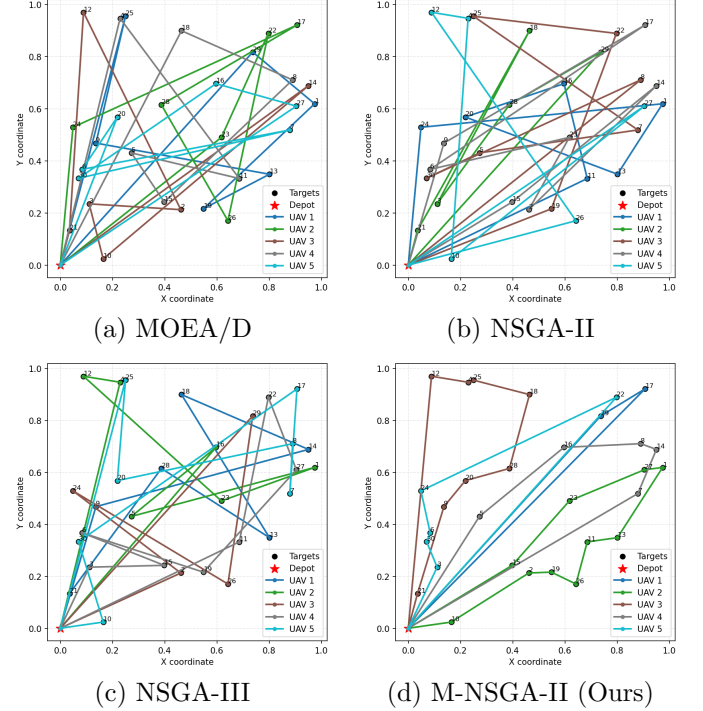


Fig. 1. Visualization of the VAR-optimal solutions found by each algorithm in Scenario S1. The paths of individual UAVs are shown in different colors.

The M-NSGA-II solution has visibly similar path lengths and fewer route intersections, supporting the quantitative finding that majorization-based ranking improves both balance and route efficiency.

4.3 Discussion

The results should be interpreted in light of the target problem class. M-NSGA-II is not intended to replace Pareto dominance for arbitrary heterogeneous MOO problems. Its advantage appears when objective components are comparable and the decision maker prefers both small magnitudes and a balanced distribution. The UAV study fits this structure because every objective is a path length assigned to one UAV. In this setting, weak majorization suppresses solutions where one UAV carries an excessive burden, while crowding distance maintains alternatives among incomparable balanced solutions.

The experiment also suggests why TD can improve even though the ranking rule is motivated by balance. Highly unbalanced assignments often induce inefficient route geometry: one agent may visit distant or poorly ordered targets while others are underused. By reducing such assignments, M-NSGA-II can focus search on route structures that are both balanced and short.

One should also distinguish the proposed method from simply adding VAR or MAX as extra objectives. A metric-augmentation approach depends on choosing a particular balance measure in advance, and optimizing one measure does not necessarily improve another. By contrast, weak majorization defines an ordering that is consistent with an entire class of monotone Schur-convex preferences. This is why the experimental evaluation reports TD, VAR, and MAX together: the aim is not to tune the algorithm to one chosen metric, but to test whether the induced ranking improves several balance-related criteria simultaneously.

5. CONCLUSION

This paper introduced Majorization-NSGA-II, an evolutionary algorithm for homogeneous multi-objective problems where balanced solutions are desired. The method modifies the ranking step of NSGA-II by replacing Pareto dominance with weak majorization. We showed that this selection rule is theoretically justified for aggregate preferences that are both Schur-convex and monotonically increasing, and we clarified the practical conditions under which this assumption is appropriate. In a multi-UAV path-planning case study, M-NSGA-II produced statistically significant improvements over MOEA/D, NSGA-II, and NSGA-III in total distance, variance, and maximum individual cost.

The present study also has limitations. The theoretical analysis provides an order-preserving justification for a class of balance-preferring objectives, but it does not establish convergence guarantees for the evolutionary process. The case study focuses on obstacle-free path planning with a moderate number of homogeneous objectives; obstacle-rich environments, broader balance-aware baselines, and very high-dimensional objective spaces should be examined in future work. These limitations notwithstanding, the results indicate that weak majorization is a useful and principled ranking mechanism for balanced optimization in homogeneous multi-component systems.

REFERENCES

- Amir, M., Bettini, M., and Prorok, A. (2025). When is diversity rewarded in cooperative multi-agent learning? *arXiv preprint arXiv:2506.09434*.
- Bakry, W., Rashid, A., Al-Mohamad, S., and El-Kanj, N. (2021). Bitcoin and portfolio diversification: A portfolio optimization approach. *Journal of Risk and Financial Management*, 14(7), 282.
- Bettini, M., Shankar, A., and Prorok, A. (2025). System neural diversity: Measuring behavioral heterogeneity in multi-agent learning. *Journal of Machine Learning Research*, 26(163), 1–27.
- Cesarone, F., Scozzari, A., and Tardella, F. (2020). An optimization-diversification approach to portfolio selection. *Journal of Global Optimization*, 76(2), 245–265.
- Chiam, S.C., Tan, K.C., and Al Mamun, A. (2008). Evolutionary multi-objective portfolio optimization in practical context. *International Journal of Automation and Computing*, 5(1), 67–80.
- Deb, K. and Jain, H. (2014). An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part i: Solving problems with box constraints. *IEEE Transactions on Evolutionary Computation*, 18(4), 577–601.
- Deb, K., Pratap, A., Agarwal, S., and Meyarivan, T. (2002). A fast and elitist multiobjective genetic algorithm: Nsga-ii. *IEEE transactions on evolutionary computation*, 6(2), 182–197.
- Deep, K. and Mebrahtu, H. (2011). Combined mutation operators of genetic algorithm for the travelling salesman problem. *International Journal of Combinatorial Optimization Problems and Informatics*, 2(3), 1–23.
- Devi, N., Dalal, S., Solanki, K., Dalal, S., Lilhore, U.K., Simaiya, S., and Nuristani, N. (2024). A systematic literature review for load balancing and task scheduling techniques in cloud computing. *Artificial Intelligence Review*, 57(10), 276.
- Ebeling, C.E. (2019). *An introduction to reliability and maintainability engineering*. Waveland Press.
- Harmati, I. and Skrzypczyk, K. (2009). Robot team coordination for target tracking using fuzzy logic controller in game theoretic framework. *Robotics and Autonomous Systems*, 57(1), 75–86.
- Jain, R.K., Chiu, D.M.W., Hawe, W.R., et al. (1984). A quantitative measure of fairness and discrimination. *Eastern Research Laboratory, Digital Equipment Corporation, Hudson, MA*, 21(1), 2022–2023.
- Kora, P. and Yadlapalli, P. (2017). Crossover operators in genetic algorithms: A review. *International Journal of Computer Applications*, 162(10).
- Kumar, P. and Kumar, R. (2019). Issues and challenges of load balancing techniques in cloud computing: A survey. *ACM computing surveys (CSUR)*, 51(6), 1–35.
- Marshall, A.W., Olkin, I., and Arnold, B.C. (2011). Inequalities: Theory of majorization and its applications. *Springer Series in Statistics*.
- Mishra, S.K., Sahoo, B., and Parida, P.P. (2020). Load balancing in cloud computing: a big picture. *Journal of King Saud University-Computer and Information Sciences*, 32(2), 149–158.
- Mo, Y., Chen, W., and Qiu, L. (2017). Coordinating flexible loads via optimization in the majorization order. 3495–3500.
- Mo, Y., Chen, W., You, K., and Qiu, L. (2024). Optimal (0, 1)-matrix completion with majorization ordered objectives. *Automatica*, 160, 111430.
- Mo, Y. and Qiu, L. (2024). iPOP with majorization ordered objectives in privacy and security. In *The 14th Asian Control Conference*.
- Wang, H., Rodriguez-Fernandez, A.E., Uribe, L., Deutz, A., Cortés-Piña, O., and Schütze, O. (2025). A newton method for hausdorff approximations of the pareto front within multiobjective evolutionary algorithms. *IEEE Transactions on Evolutionary Computation*, 29(5), 2104–2115.
- Zhang, Q. and Li, H. (2007). Moea/d: A multiobjective evolutionary algorithm based on decomposition. *IEEE Transactions on evolutionary computation*, 11(6), 712–731.